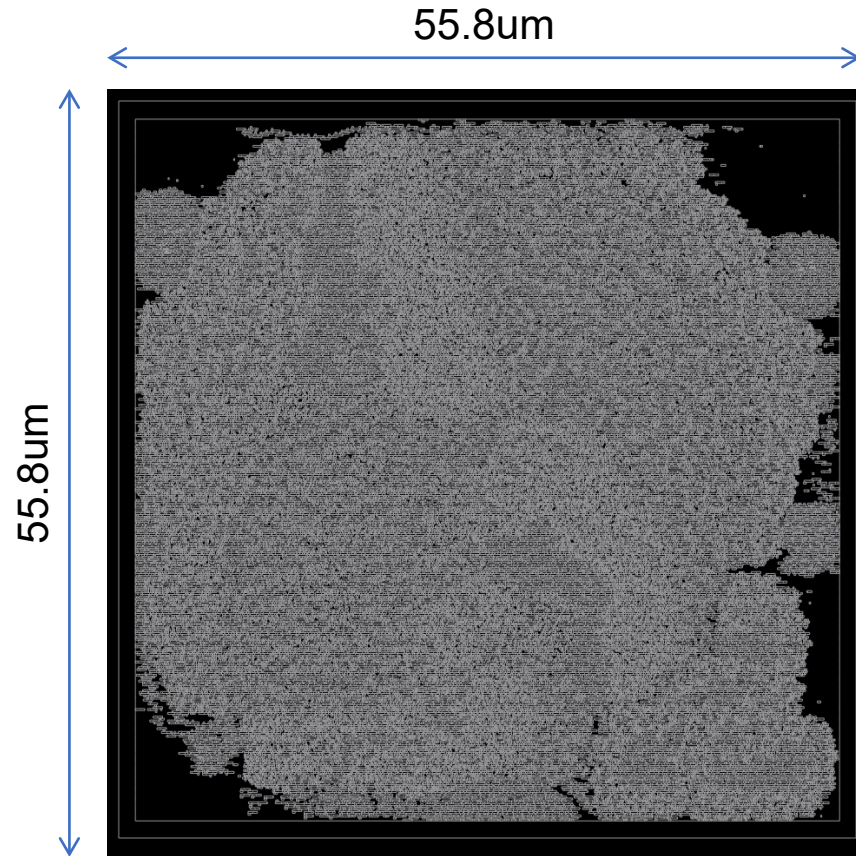
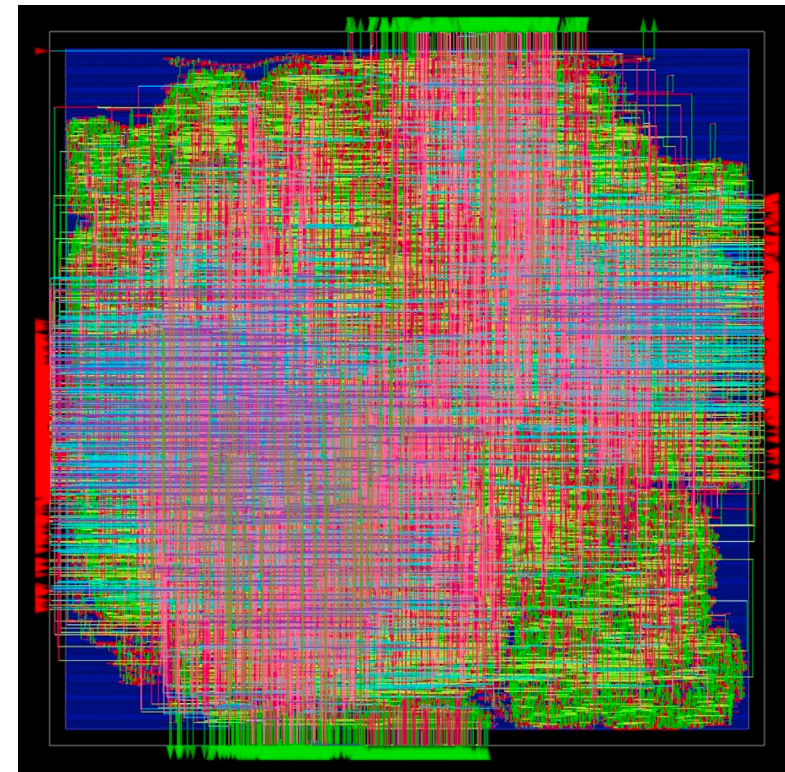


Benchmark: ECG (62K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD



Placement



Routing

Benchmark: ECG (62K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

Performance	Target freq (GHz)	2.00
	Effective freq (GHz)	1.99
	WNS (ps)	-1.30
	TNS (ps)	-23.97
Power	Total power (mW)	22.01
	Internal power (mW)	16.64
	Switching power (mW)	5.35
	Leakage power (mW)	0.03
Area	Chip dimension (um)	55.8 x 55.8
	Macro Count	0
	Cell Count	61,585
	Placement Density	72%

Placement (min)	0.63
CTS (min)	0.12
Routing (min)	13.9
Total (min)	14.7

Turnaround time

OS	Red Hat Linux 8.10
CPU	Xeon Gold 6454S
#Cores	64
CPU Speed	3400 MHz
Memory	512 GB

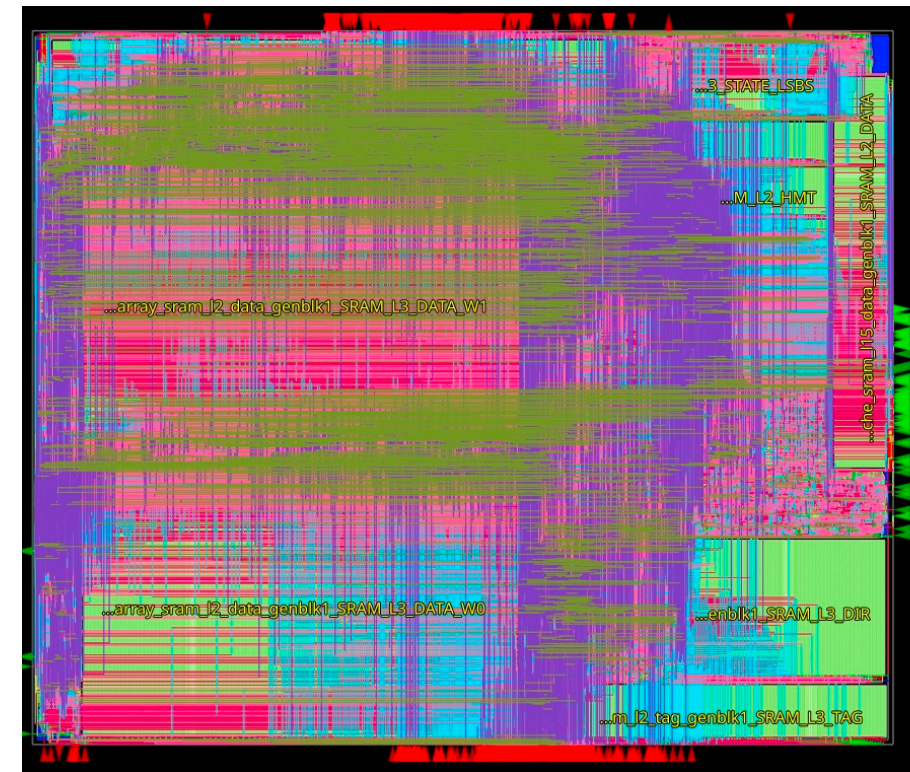
Server spec

Benchmark: OpenPiton (292K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD



Placement



Routing

Benchmark: OpenPiton (292K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

Performance	Target freq (GHz)	1.00
	Effective freq (GHz)	0.95
	WNS (ps)	-54.34
	TNS (ps)	-258.07
Power	Total power (mW)	101.87
	Internal power (mW)	94.75
	Switching power (mW)	6.39
	Leakage power (mW)	0.72
Area	Chip dimension (um)	330.0 x 398.2
	Macro Count	28
	Cell Count	292,047
	Placement Density	72%

Placement (min)	9.5
CTS (min)	2.1
Routing (min)	32.4
Total (min)	43.9

Turnaround time

OS	Red Hat Linux 8.10
CPU	Xeon Gold 6454S
#Cores	64
CPU Speed	3400 MHz
Memory	512 GB

Server spec